

# Ebullient Securities Idea Summary

Team 67

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## I. INTRODUCTION

This idea summary highlights the work completed after midterm and how the strategy evolved into its final form. The midterm phase provided valuable insights into how our strategies performed in practice and identified areas needing refinement. We gained a clearer understanding of EBX and EBY behavior across different market conditions and pinpointed where our modeling could be more stable and reliable. A key learning was observing system behavior during choppy or directionless markets. Several indicators and statistical models behaved differently depending on the regime, sometimes producing mixed signals. This highlighted the need for better signal filtering and confirmation, and emphasized matching momentum cues with reliable trend direction rather than relying on single-layer logic. Another important outcome was recognizing the need for custom indicators. While existing transforms performed well, they missed smaller structural changes and sudden regime shifts. We designed new proprietary indicators to smooth noisy behavior, improve cross-regime consistency, and capture patterns previously overlooked. Based on these learnings, we implemented several post-midterm updates: improved indicator calculation stability, added clearer trade confirmation steps, refined volatility-based activation rules to avoid low-quality periods, and made exit logic more adaptive to handle both slow and fast market movements. These changes made the strategy more robust, consistent, and aligned with real trading conditions.

## II. MARKET STRUCTURE ANALYSIS

Our data analysis consisted of two complementary studies aimed at understanding the underlying structure of the intraday price series. First, we performed a statistical analysis of the early-day mean and volatility to evaluate how initial market conditions influence intraday behaviour and regime formation. This allowed us to identify shifts in market state and quantify the strength of expansion or compression regimes. Second, we conducted a detailed examination of multiple moving-average based features to assess their responsiveness, smoothness, and ability to filter choppy or low-quality trading environments. By combining these two perspectives, we established a clearer framework for identifying reliable trading windows and avoiding structurally unfavourable market conditions.

### A. Statistical Analysis

We conducted an extensive statistical study of the intraday price behaviour across all trading days of EBX and EBY, in order to understand how early dynamics influence the remainder of the day. Our analysis focused on the first 30 minutes of each session and evaluated two primary quantities: the early-day mean price  $\mu_{0-30}$  and the early-day volatility

$\sigma_{0-30}$ , both computed relative to the daily reset price of 100.

To ensure robustness, we experimented with multiple window lengths (5, 15, 30, 45, and 60 minutes). Shorter windows suffered from high noise and poor predictive stability, particularly when estimating  $\sigma$ , while longer windows delayed signal availability and significantly reduced the number of usable trading opportunities. The 30-minute window provided the strongest balance between predictive power and practical trade frequency.

**Mean (Directional) Behaviour:** The early-day mean  $\mu_{0-30}$  indicated that the market had shifted from its opening state, but this shift was not strongly directional. In the case of EBY for example, although 64.7% of days with  $\mu_{0-30} > 100$  closed higher, and 46.5% of days with  $\mu_{0-30} < 100$  still closed above 100, the separation was not decisive. We therefore interpreted  $\mu$  primarily as a signal of *market state change* rather than a reliable predictor of final direction, and designed conditional rules to exploit this state shift.

**Volatility (Regime) Behaviour:** Early-day volatility provided an orthogonal and equally important dimension. High- $\sigma$  days (above the 75th percentile) exhibited significantly larger peak-to-peak intraday excursions, often exceeding 1.0 in magnitude. Low- $\sigma$  days remained confined to narrow ranges and displayed mixed or mean-reverting behaviour. Crucially, combining  $\mu$  and  $\sigma$  produced four distinct predictive regimes. High  $\mu$  with high  $\sigma$  yielded the strongest continuation behaviour (82% of such days closed above 100), whereas low  $\mu$  with high  $\sigma$  produced the strongest downside trend (71.5% closed below 100). Low- $\sigma$  regimes showed weaker and noisier outcomes regardless of direction. High volatility in 1 to 5 min period in EBX on average showed high unpredictably rest of the day.

**Overall Market Structure:** EBX exhibits a two-factor intraday structure: (1) early direction determines drift, and (2) early volatility determines excursion magnitude. The combination forms a stable and statistically separable regime system that informs the design of our strategies.

TABLE I: Overall 30-Minute Statistical Summary for EBX and EBY

| Metric                     | EBX      | EBY      |
|----------------------------|----------|----------|
| Overall Mean $\mu$         | 100.0153 | 100.0017 |
| Overall Std. Dev. $\sigma$ | 0.5353   | 0.5336   |
| $\sigma$ (50th percentile) | 0.07     | 0.08     |
| $\sigma$ (75th percentile) | 0.11     | 0.12     |
| Avg. Price at 30 min       | 100.01   | 99.99    |
| Avg. Close of Day          | 100.02   | 100.01   |
| Avg. (High – Price@30)     | 0.47     | 0.49     |
| Avg. (Price@30 – Low)      | 0.52     | 0.49     |

The following metrics summarize the aggregate behaviour of EBX and EBY over all sessions and highlight the statistical structure that justified our regime-based approach.

## B. Novelty Framework

### 1) Objective

Direct analysis on the noisy and granular data of price gave too many false signals to be directly incorporated, making them unfit for trading. To address this, we applied a range of moving averages on the price data. Each moving average was assessed based on its ability to smoothen the data, lag characteristics, responsiveness, and remove seasonality. This analysis helped us identify which moving average was the most balanced in terms of noise reduction and timely response to market movements.

### 2) Analysis

To smooth out the granular data, we applied a range of established moving average techniques and analyzed their behaviour across different market conditions. Here are a few of the moving averages we analysed:

**Jurik Moving Average (JMA)** was really impressive in terms of smoothness and effectiveness and filtering out short term noise with lower lag compared to its other alternatives. Its lack of responsiveness to sharp market moves was a significant drawback.

**Ehler's Super Smoother (ESS)** excelled at smoothening the trend with low lag, but its primary drawback was its extreme responsiveness during choppy periods, which made it unusable directly.

**Kaufman's Adaptive Moving Average (KAMA)** had the highest smoothening but with the most lag and lacked responsiveness. It could be used as a good confirmation filter, as it filtered out the choppy markets really well.

Amongst the existing methods, the **Zero Lag Exponential Moving Average (ZLEMA)** provided the most balance between responsiveness, lag reduction and smoothness. Nevertheless, it did overreact to choppy markets which diminished the overall robustness. Each of these moving averages had its own strengths, but none of them had the balance between smoothening, responsiveness, lag and seasonality reduction that we wanted.

### 3) Chaos to Clarity

To address the limitations observed in the existing moving averages, we branched away from the existing moving averages to create our own custom indicator.

Introducing **AURA (Adaptive Ultra-smooth Responsive Average)** - our very own custom indicator which blends the smoothening logic of KAMA with the smoothening and

responsive capabilities of Ehler's Super Smoother to give a perfect balance between smoothening and responsiveness. Its key highlight is that it reacts to sharp spikes yet still shows the overall trend of the market by removing the seasonality of the trend. AURA uses the Ehler's Super Smoother output as its input:

$$x_t = \text{SuperSmoother}(t).$$

The Efficiency Ratio over period  $P$  is

$$ER_t = \frac{|x_t - x_{t-P}|}{\sum_{i=t-P+1}^t |x_i - x_{i-1}|}.$$

Fast and slow smoothing constants are

$$\alpha_f = \frac{2}{F+1}, \quad \alpha_s = \frac{2}{S+1},$$

where  $F$  is the fast period (high responsiveness) and  $S$  is the slow period (high smoothness).

The adaptive factor is

$$SC_t = (ER_t(\alpha_f - \alpha_s) + \alpha_s)^2.$$

The recursive update rule is:

$$\text{AURA}_t = \text{AURA}_{t-1} + SC_t(x_t - \text{AURA}_{t-1})$$

with base case  $\text{AURA}_0 = x_0$ .



Fig. 1: AURA indicator visualization

We took this even further by applying the same smoothening on many different moving averages like JMA, ZLEMA, etc. but none of them were as effective as AURA.

To further improve on the false choppy crossovers, we needed a more refined smoothener when the market is specifically choppy. We needed a hybrid of responsiveness during trendy regions to reduce lag and smoothness to reduce false signals during choppy market. To do this we created an even better indicator - **HYDRA (Hybrid Regime Aware) Moving**

**Average** which blends AURA with its longer lookback version  $AURA_{slow}$  using a volatility-driven regime weight. We calculate the rolling standard deviation of the rolling standard deviation of the slope of price to distinguish trendy regions (with high peaks) and choppy regimes (lots of small values). We then compute this value for the first hour of the day and find out its 99th percentile value. We then divide it by 2.5 to get the threshold for that entire day which will help us divide the day into trendy and choppy regimes using the data of the first hour.

$$b_t = \begin{cases} 1, & \text{if } \text{Diff\_STD\_STD}_t > \text{STD\_Threshold}, \\ 0, & \text{otherwise.} \end{cases}$$

For a window length  $W$ ,

$$c_t = \sum_{i=t-W+1}^t b_i, \quad w_t = \frac{c_t}{W}.$$

Let  $A_t$  denote AURA and  $S_t$  denote  $AURA_{slow}$ . HYDRA is computed as

$$\text{HYDRA}_t = w_t A_t + (1 - w_t) S_t.$$

“““During the first 60 minutes,

$$\text{HYDRA}_t = S_t.$$



Fig. 2: HYDRA indicator visualization

“““

### III. INVESTIGATED METHODOLOGIES

#### A. AURA-KAMA Crossover

The AURA-KAMA Crossover Strategy enters long when AURA crosses above KAMA with SMA calculated on Heikin Ashi candles for confirmation. And enters short when AURA crossover below KAMA with SMA confirmation. The problem with this strategy was it gave a lot of signals during choppy

regimes where ideally we would want to avoid taking any signals.

#### B. Temporal Fusion Transformer (TFT)

To effectively capture the non-linear dynamics and complex dependencies inherent in the time-series data, we deployed a Temporal Fusion Transformer (TFT). This attention-based architecture is specifically designed for interpretable multi-horizon forecasting. The model is optimized to process a high-dimensional feature space, utilizing a lookback window of 120 time steps populated with significant features extracted via XGBoost.

##### a) Architecture

The TFT architecture synthesizes several specialized components to mitigate noise and extract temporal patterns:

- **Variable Selection Network (VSN):** Incorporates Gated Residual Networks (GRNs) to dynamically weigh inputs, effectively isolating relevant features while suppressing noise from less predictive indicators.
- **Local Processing:** A 3-layer LSTM encoder with a hidden size of 128 is employed to encode local trends and capture immediate short-term temporal interactions.
- **Temporal Fusion Decoder:** This module utilizes a Multi-Head Attention mechanism (8 heads) to model long-range dependencies across the entire lookback window, complemented by a position-wise Feed-Forward Network (hidden size 256).
- **Gated Output:** A Gated Linear Unit (GLU) is applied to regulate the information flow into the final output layer, ensuring precise control over the prediction signal.

##### b) Trading Strategy

The trading strategy leverages the probabilistic nature of the TFT’s output to generate high-confidence signals. By enforcing a strict confidence interval, the system initiates a long position only when the predicted 10th percentile (P10) exceeds the minimum profit threshold, and a short position when the 90th percentile (P90) drops below the negative threshold. This logic ensures a 90% confidence level for favorable returns before entry. Risk is managed via a fixed bracket order comprising a stop-loss and a take-profit, alongside a mandatory end-of-day square-off.

#### C. BlackFlame Reversion

The BlackFlame Reversion strategy trades the EBY dataset using a multi-confirmation entry process and a three-level exit framework. The core idea is to take positions only when mean-reversion momentum and Jurik Moving Average (JMA) based trend conditions align. Momentum is derived from a Z-score transformed through a sigmoid function, allowing the system

to detect when price deviates significantly from its short-term mean. Trend direction is validated using three JMAs (fast, mid, and slow). A day becomes eligible for trading only when volatility, measured by ATR, exceeds a minimum threshold.

When a momentum signal appears, the strategy does not enter immediately. Instead, it marks the signal as “pending” and waits for a short confirmation window. During this window, the JMAs must show a clear upward slope for a long entry or a clear downward slope for a short entry. This prevents reacting to short-term noise and ensures that the mean-reversion impulse is supported by broader directional context. If the trend confirmation does not occur within the allowed time, the signal expires and no trade is taken.

The exit mechanism follows a structured three-level TP/SL (take-profit / stop-loss) design. Initially, the system checks for an absolute take-profit or stop-loss. As the trade moves into profit, the stop is shifted toward a break-even level, ensuring no further downside risk. Once profit expands further, two stages of trailing-stop tightening occur: the first allows moderate flexibility, while the second is more restrictive, securing profits more aggressively as the move extends. The trailing logic follows the best price reached, preserving gains while allowing upward continuation. Any position still open at the end of the trading day is closed.

#### IV. MAIN STRATEGY: RAGNARÖK TREND PROTOCOL

##### A. Objective and Day Segregation

The core production system employs a dynamic selector which, on each day, activates the strategy expected to yield the highest net pnl return after transaction costs. In the mid-term stage, We filtered out the most volatile tradable days based on ATR and STD threshold values over the whole day. This yielded 60 high-opportunity days in EBY and 101 such days in EBX.

In the final stage, we augmented the segregation framework by incorporating information from the first 30 minutes of the session. We calculated the threshold values of mean and standard deviation over the whole dataset and compared with the calculation of first 30 min. Using percentile values of standard deviation we segregated the days as follow:

- 1)  $\mu_{0-30} > 100$  and  $\sigma_{0-30} > \sigma_{75}$ ,
- 2)  $\mu_{0-30} > 100$  and  $\sigma_{50} \leq \sigma_{0-30} \leq \sigma_{75}$ ,
- 3)  $\mu_{0-30} < 100$  and  $\sigma_{0-30} > \sigma_{75}$ ,
- 4)  $\mu_{0-30} < 100$  and  $\sigma_{50} \leq \sigma_{0-30} \leq \sigma_{75}$ ,

where  $\sigma_{50}$  and  $\sigma_{75}$  denote the 50th and 75th percentiles of the 30-minute volatility distribution.

##### B. Sub-Strategies

At runtime, the manager evaluates signals based on priorities and locking mechanisms to generate final signals.

##### 1) Denoised Trend Pattern:

This candidate uses Kalman Filter and Unobserved Component Model crossover for identifying winning trends. A long window KAMA Filter is used for confirmation as it filters out low-SNR regimes. Logic computes KF trend, UCM and KAMA on a core price-based feature. Enters long when KF is less than equal to UCM and short when KF is greater than equal to UCM and absolute KAMA slope greater than 0.001.

**Kalman Filter** is a recursive algorithm for estimating the state of a dynamic system from noisy measurements. It provides smooth trend estimates that filter out measurement noise while quickly adapting to genuine changes in price direction.

**Unobserved Components Model (UCM)** decomposes a time series into latent level and trend components that evolve recursively over time. It updates these components using exponentially weighted adjustments controlled by parameters  $\alpha$  and  $\beta$ , allowing the model to capture gradual structural changes while suppressing transient noise. This yields a smoothed, causal trend estimate that reacts adaptively to genuine price shifts without introducing forward bias.

**Kaufman’s Adaptive Moving Average (KAMA)** is a dynamic smoothing filter that adjusts its sensitivity based on the balance between directional movement and total volatility. When price changes are consistent, KAMA becomes more responsive; when noise dominates, it smooths more aggressively. This adaptive behavior makes KAMA effective for distinguishing genuine directional moves from random fluctuations, producing stable yet timely trend estimates.

##### 2) Sigma-Triggered Alpha Reactor (STAR):

This system is an **adaptive framework** designed to select the optimal trading logic and directional bias based on the asset’s **initial market volatility** ( $\sigma$ ) measured during the first 30 minutes of the trading day. This approach ensures optimal risk controls and directional alignment for the prevailing market regime.

##### a) Adaptive Strategy Selection

The trading day is dynamically categorized into two regimes using thresholds derived from the historical distribution of 30-minute opening volatility:

If the day’s volatility or mean price falls outside these profitable regimes, the system remains flat.

##### b) Core Sequential Trade Logic (Gap-Fill and Reversal)

Both strategies utilize a **sequential, reversal-based structure**, adapted to their specific bias, involving three distinct phases. These phases have been set based on the overall mean, as the overall mean of the first 30 minutes of the whole dataset

TABLE II: Volatility Regimes and Associated Trading Strategies

| Regime                                 | Volatility Condition                                                          | Bias & Role                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Strategy 1</b><br>(High Volatility) | $\sigma > \sigma_{75}$ and $\mu > \mu_{\text{threshold}}$                     | <b>Short-First</b><br>Assumes price spike/gap up is likely to be sold off.  |
| <b>Strategy 2</b><br>(Mid Volatility)  | $\sigma_{50} \leq \sigma \leq \sigma_{75}$ and $\mu < \mu_{\text{threshold}}$ | <b>Long-First</b><br>Assumes price drop/gap down is likely to be bought up. |

is almost same as initial capital, so for mid to high volatility days if mean price of first 30 minutes of the day is less than initial capital then it was found that if price goes down by some threshold then there is a prediction of covering that down threshold or go higher to maintain mean and similarly opposite behaviour for the first 30 minutes mean greater than 100 of other days:

- 1) **Gap-Fill Entry:** The trade starts with an initial, counter-trend position taken at the 30-minute mark if the price is outside a specific threshold ( $\text{Open} \pm \text{Threshold}$ ). This aims to profit from the immediate filling of the initial gap.
- 2) **Flip:** When the gap-fill trade hits its target, the system immediately **flips** (closes the initial position, waits a brief cooldown period, and opens the main, high-conviction position in the favored direction) with the take profit along with the trailing stop loss.
- 3) **Position Management:** The main position uses unique **trailing stops** and a **reversal target** to capture profit and establish a secondary reversal trade or square-off at the end of the day.

### 3) Volatility Filtered KAMA Trend:

This strategy first performs day segregation using ATR and STD thresholds; once both thresholds are met, the day is declared tradable. On these filtered high-quality tradable days, it applies the Kaufman Adaptive Moving Average (KAMA) filter slope to detect trend direction and accordingly generates long or short signals. For exits, if it is the last tick of the day, the system squares off while maintaining the minimum trade duration. Otherwise, it sets a take-profit barrier; if this barrier is crossed, the current signal exits and the system resets for the next signal while maintaining the minimum cooldown duration.

### 4) Hawkes Adaptive Trend Strategy:

This strategy is built to capture **large, sustained price movements** while avoiding choppy and low-quality market conditions. Trading decisions are made on a denoised, adaptive

trend rather than raw price.

#### a) Signal and Entry Logic

- 1) **Adaptive Trend:** Price is smoothed using the **Ehlers Super Smoother** followed by the **AURA** filter, producing a trend line that adapts dynamically to market conditions.
- 2) **Impulse Detection (Hawkes):** A **Hawkes Process** measures the intensity of recent momentum bursts to distinguish genuine impulses from noise.
- 3) **Dynamic Volatility Walls:** Adaptive volatility-based boundaries are constructed to represent evolving support and resistance levels.
- 4) **Entry Condition:** Trades are initiated only when a validated Hawkes impulse interacts with a volatility wall and the underlying trend slope confirms direction.

#### b) Risk Controls

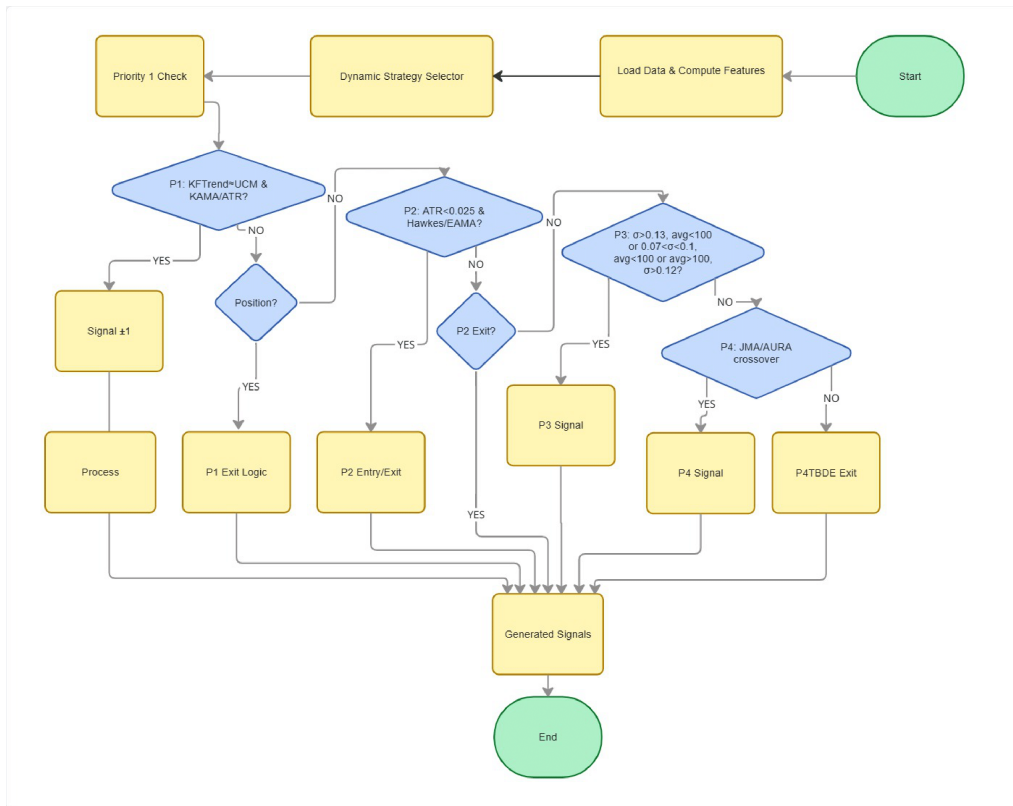
- **ATR Volatility Gate:** Prevents trading during extremely low-volatility regimes.
- **Aggressive Trailing Stop:** Rapidly secures unrealized gains and minimizes drawdowns during reversals.

#### 5) SuperNova Crossover

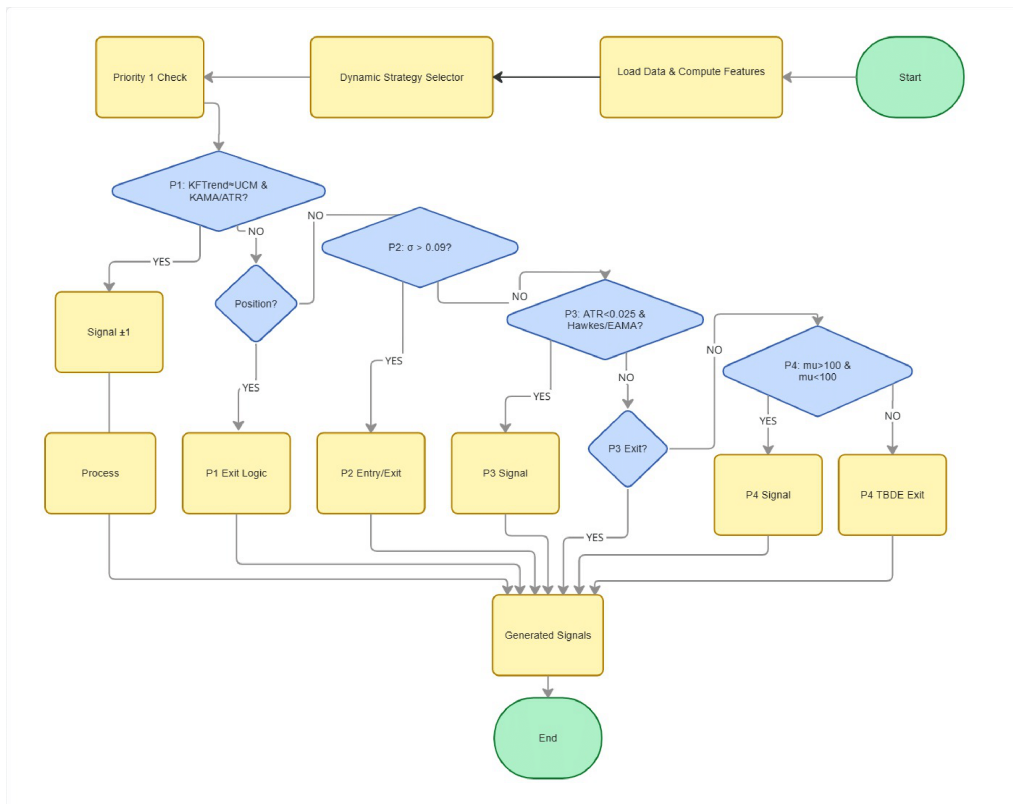
This strategy implements a volatility gated AURA-JMA crossover that only activates when the market shows sufficient movement. A day becomes tradable only when both ATR and rolling standard deviation conditions are met. Once activated, it trades based on the crossovers between a slower version of AURA and the Jurik Moving Average. It also has exits based on take profits and trailing stop losses. Signals are treated at an independent level which means any exit signal also activates a delayed entry in the opposite direction from the previous trade.

#### C. Dynamic Priority Based Strategy Manager

For selecting the most optimal strategy, we use a priority based locking method. We have pre-assigned priorities to each strategy and each strategy lock resets every day.



(a) EBX Strategy Flowchart



(b) EBY Strategy Flowchart

## 1) *EBX*

The selector dynamically prioritizes three strategies: **Denoised Trend Pattern**, **Volatility-Filtered KAMA Trend**, and **Sigma-Triggered Alpha Reactor (STAR)** with primary emphasis on the Denoised Trend Pattern. **Denoised Trend Pattern** stabilizes direction by filtering noise and preventing micro-volatility-driven false signals. If no valid signal is found, the system evaluates the **Volatility-Filtered KAMA Trend**, which uses ATR thresholds to confirm clean trend or mean-reversion phases in moderate to high-volatility conditions. Finally, **STAR** assesses low to mid-volatility environments by comparing early-session standard deviation and mean values with overall market statistics. Optimized on EBX behavior, these layers collectively adapt to stable trends, volatility shifts, and transitional regimes, ensuring the most reliable strategy is selected for each condition.

## 2) *EBY*

The selector prioritizes four strategies: **Denoised Trend Pattern**, **Sigma-Triggered Alpha Reactor (STAR)**, **Volatility-Filtered KAMA Trend**, and **Money Heist** with primary weight on the **Denoised Trend Pattern**. This first priority filters noise and stabilizes direction, allowing reliable identification of strong trend-following setups, especially on high-volatility days. **STAR** then evaluates high-to-mid volatility conditions by ranking days using early-session standard deviation and mean against full-day thresholds. The **Volatility-Filtered KAMA Trend** targets mid-to-low volatility phases by applying ATR-based criteria to validate cleaner mean-reversion or controlled-trend environments. Finally, the **Money Heist** strategy derives direction from the first 30-minute mean relative to overall mean thresholds, entering long or short accordingly, applying a defined take-profit, locking 5%, and activating a trailing stop to maximize continuation gains. Together, these layers adapt across trend regimes and volatility states, enabling the selector to consistently choose the most suitable strategy for prevailing market conditions

## V. METRICS

TABLE: Performance Metrics for EBY and EBX

| Metric             | EBY    | EBX    |
|--------------------|--------|--------|
| Final Returns      | 34.75% | 39.68% |
| Annualized Returns | 31.39% | 20.41% |
| Calmar Ratio       | 15.94  | 10.36  |
| Annualized Calmar  | 14.40  | 5.33   |
| Maximum Drawdown   | -2.18% | -3.83% |
| Sharpe Ratio       | 2.58   | 2.48   |

## VI. CONCLUSION

### A. Key Findings

This research successfully extended the multi-regime quantitative trading framework with advanced statistical models and new trading strategies:

- **HYDRA Strategy:** A multi-signal system that combines multiple indicators to identify high-probability trading opportunities.
- **Hawkes Process Mean Reversion Bands:** Dynamic bands that adapt to volatility clustering, helping identify when prices are likely to revert to their mean.
- **Z-Score Analysis:** Statistical method for measuring price deviations, enabling data-driven position sizing and risk assessment.
- **Buy-and-Hold Opportunity Detection:** Framework for identifying market conditions where holding positions outperforms active trading, improving capital efficiency.

### B. Implementation Insights

- **Custom Indicators:** Developed proprietary technical indicators designed specifically for different market conditions and strategy needs.
- **Enhanced Features:** Expanded beyond traditional indicators to include volume analysis, volatility metrics, and market microstructure data.
- **Risk Management:** Implemented Z-score-based position sizing and cascading controls that automatically adjust exposure based on market conditions.